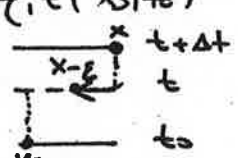


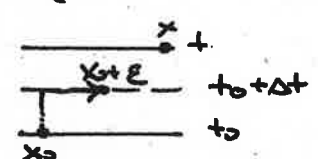
Alternative treatment: Fokker Planck backwards equation.

quick review: FOKKER PLANCK

Chapman Kolmogorov Equation

$$(I) \underbrace{P(x, t+\Delta t | x_0, t_0)}_{\text{fixed initial condition}} = \int_{-\infty}^{\infty} d\xi \underbrace{P(x, t+\Delta t | x-\xi, t)}_{t_0 < t < t+\Delta t} P(x-\xi, t | x_0, t_0)$$


or: backwards

$$(II) \underbrace{P(x, t | x_0, t_0)}_{\text{fixed state}} = \int_{-\infty}^{\infty} d\xi \underbrace{P(x, t | x_0+\xi, t_0+\Delta t_0)}_{\text{varying initial condition}} P(x_0+\xi, t_0+\Delta t_0 | x_0, t_0)$$


- Kramers Moyal Expansion of (I) with $f(x) = P(x+\xi, t+\Delta t | x, t) P(x, t | x_0, t_0)$ and $f(x-\xi)$ Taylor expansion yields FORWARD FOKKER PLANCK.

ie:
$$\partial_t P(x, t | x_0, t_0) = -\partial_x (A(x) \cdot P(x, t | x_0, t_0)) + \frac{1}{2} \partial_x^2 (D(x) P(x, t | x_0, t_0))$$

initial condition

$$P(x, t=t_0 | x_0, t_0) = \delta(x-x_0)$$

- One can repeat the Kramers Moyal Expansion for BACKWARD Chapman-Kolmogorov: (II)

$$\partial_{t_0} P(x, t | x_0, t_0) = -A(x_0) \frac{\partial}{\partial x_0} P(x, t | x_0, t_0) - \frac{1}{2} D(x_0) \frac{\partial^2}{\partial x_0^2} P(x, t | x_0, t_0)$$

BACKWARD FOKKER PLANCK EQ.

note: sign change and operator act only on $P(x, t | x_0, t_0)$

For a homogeneous process:

$$P(x, t | x_0, t_0) = P(x, 0 | x_0, t_0 - t)$$

it follows that:

$$\partial_t P(x, t | x_0, t_0) = \partial_{t_0} P(x, t | x_0, t_0) = -\partial_t P(x, 0 | x_0, t_0 - t)$$

Note: the backward I.P. can be used for escape rate problems (next)

Alternative Derivation of the FOKKER-PLANCK EQUATION using ITO's LEMMA

Reminder ITO's LEMMA:

$f(x)$ where f is a function of x , a stoch. Markov process:

$$dx = A(x,t)dt + \underbrace{B(t)dt^{1/2}}_{\equiv dW(t) \text{ Wiener Increment}} \quad B(t) = \lim_{dt \rightarrow 0} N(0, dt)$$

Proof on Equivalence of S.D.E and Fokker Planck P.D.F. Eq.:

Ito's Lemma reminder:

$$df = (A(x,t) \frac{df}{dx} + \frac{1}{2} B(x)^2 \frac{d^2f}{dx^2}) dt + B(x) dW$$

Next: $p(x_0, t_0 | x, t)$ obeys:

$$\langle f(x(t)) \rangle = \int dx f(x) p(x_0, t_0 | x, t)$$

and:

$$\begin{aligned} \frac{d \langle f(x(t)) \rangle}{dt} &= \int_{-\infty}^{\infty} dx \frac{d}{dt} (f(x) p(x_0, t_0 | x, t)) \\ &= - \int_{-\infty}^{\infty} dx f(x) \frac{d}{dt} p(x_0, t_0 | x, t) \end{aligned}$$

and:

$$\begin{aligned} \frac{d \langle f(x(t)) \rangle}{dt} &= \langle \frac{df(x(t))}{dt} \rangle = \int_{-\infty}^{\infty} \underbrace{\langle \frac{df(x(t))}{dt} \rangle}_{\text{ITO LEMMA}} \cdot p(x_0, t_0 | x, t) dx \\ &= \int_{-\infty}^{\infty} \left[\underbrace{A(x,t) \frac{df}{dx}}_{\text{1st Term}} + \underbrace{\frac{1}{2} B(x)^2 \frac{d^2f}{dx^2}}_{\text{2nd Term}} \right] p(x_0, t_0 | x, t) dx \end{aligned}$$

Integrate by parts:

alternative: replace $f(x) = \int_{-\infty}^x \tilde{f}(x) dx$ and use $\int \frac{d}{dx} \tilde{f}(x) f(x) dx = \frac{df}{dx} \tilde{f}$ $\rightarrow$ count IBP.

$$\begin{aligned} \text{1st Term: } \int_{-\infty}^{\infty} \underbrace{A(x,t)}_u \underbrace{\frac{df}{dx}}_{v'} p(x_0, t_0 | x, t) dx &= \int_{-\infty}^{\infty} \underbrace{f(x)}_u \underbrace{\frac{d}{dx} [A(x,t) p(x_0, t_0 | x, t)]}_{v'} dx \\ &\quad - \underbrace{[A(x,t) p(x_0, t_0 | x, t)] f(x)}_{\text{vanishes (0)}} \bigg|_{-\infty}^{\infty} \quad \text{Surface Term} \\ &\quad \lim_{x \rightarrow \infty} p(x_0, t_0 | x, t) = 0 \end{aligned}$$

2nd Term

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{2} B(x)^2 \frac{d^2f}{dx^2} p(x_0, t_0 | x, t) dx &= -\frac{1}{2} \int \frac{d}{dx} [B(x)^2 p(x_0, t_0 | x, t)] \frac{df}{dx} dx \\ &\quad + \underbrace{p(x_0, t_0 | x, t) B(x)^2}_{\text{Surface Term} \rightarrow 0} \bigg|_{-\infty}^{\infty} \end{aligned}$$

2nd Int. by parts

$$= \int \frac{1}{2} \frac{d^2}{dx^2} [B(x)^2 p(x_0, t_0 | x, t)] f(x) dx$$

$$\Leftrightarrow \int dx f(x) \frac{d}{dt} p(x_0, t_0 | x, t) = \int dx f(x) \left[\frac{d}{dx} (A(x,t) p(x_0, t_0 | x, t)) + \frac{1}{2} \frac{d^2}{dx^2} (B(x)^2 p(x_0, t_0 | x, t)) \right]$$

$$\Leftrightarrow \boxed{\frac{d}{dt} p(x_0, t_0 | x, t) = - \frac{d}{dx} [A(x,t) p(x_0, t_0 | x, t)] + \frac{1}{2} \frac{d^2}{dx^2} [B(x)^2 p(x_0, t_0 | x, t)]} \quad \text{F.P. Eq.}$$

Lecture 5

- Meta-stability -

Reference

C. Gardiner
"Stochastic Methods"

Aim of lecture is to address Brownian Motion in an asymmetric well where particle can escape.

→ relevant to many processes in Physics (BTE, ...) & Chemistry (reactions ...)



$\langle T \rangle$ escape time (mean) i.e. FIRST PASSAGE TIME

Two approaches: I) Perturbation theory & II) backward F.P. Eq.

I. Stationary solution of the homogeneous F-Plank equation

$$A(x, t) = A(x)$$

$$D(x, t) = D(x)$$

$$\Rightarrow P(x, t | x_0, t_0) = P(x, t - t_0 | x_0, 0)$$

Stationary solutions $P_S(x | x_0)$:

$$\frac{dP_S}{dt} = \frac{\partial}{\partial x} [A(x) P_S(x)] - \frac{1}{2} \frac{d^2}{dx^2} [D(x) P_S(x)] \stackrel{!}{=} 0$$

can be rewritten in terms of a probability current:

$$\frac{\partial P(x, t)}{\partial t} = -\frac{\partial}{\partial x} J(x, t) \quad \wedge \quad J(x, t) = A(x) P_S(x) - \frac{1}{2} \frac{\partial}{\partial x} [D(x) P_S(x)]$$

$$\text{continuity equation i.e. } \frac{\partial P}{\partial t} + \frac{\partial J}{\partial x} = 0.$$

comment on the boundary conditions:

Steady state solutions take the form $J(x, t) = J(x) = 0 \stackrel{!}{=} \frac{\partial P_S}{\partial t}$

thus $J = \text{const.}$ since $0 = \frac{\partial P_S}{\partial t} = -\frac{\partial}{\partial x} J(x, t)$

- If we treat problem on a finite interval $x \in [a, b]$ then $J(a) = J(b) = J = \text{const.}$
- If boundary is reflecting (e.g. $x=a$) then $J(a)=0$ (i.e. no current)
 $\Rightarrow J=0 \forall x.$
- Potential solution ($J=0$ solution).

$$J = A(x) P_S(x) - \frac{1}{2} \frac{d}{dx} [D(x) P_S(x)] \stackrel{!}{=} 0$$

$$\Leftrightarrow \underbrace{\frac{\partial}{\partial x} [D(x) P_S(x)]}_{\equiv u(x)} = 2 A(x) P_S(x) \hat{=} \left(\frac{2 A(x)}{D(x)} \right) \cdot \underbrace{(D(x) P_S(x))}_{\hat{=} u(x)}$$

Stat. Phys IV
Lecture 5
Metastability

$$\Leftrightarrow \log(u(x)) = 2 \int_0^x \frac{A(x')}{D(x')} \cdot dx'$$

$$\Leftrightarrow D^{1/2}(x) \cdot P_S(x) = \exp\left(2 \cdot \int_0^x \frac{A(x')}{D(x')} dx'\right)$$

hence

$P_S(x) = \frac{N_0}{D^{1/2}(x)} \cdot \exp(-\Phi(x))$ $\Phi(x) = -2 \int_0^x dx' \frac{A(x')}{D(x')}$	(POTENTIAL solution) N_0 : normalization. Potential function.
--	--

e.g. For the smoluchowski equation we have: $D = \text{const} = (m\gamma / k_B T)$

thus $-2 \int_0^x \frac{F(x')}{D} dx' = \frac{V(x)}{k_B T}$ $\boxed{\Phi(x) = \frac{V(x)}{k_B T}}$ $A(x) = \frac{F(x)}{m\gamma}$ force on particle

- Example: diffusion in a gravitational field: (overdamped particle)

$F = -m \cdot g$, $D = \frac{k_B T}{m\gamma}$ $dx = -g dt + \sqrt{D} \cdot dW(t)$ Stochastic Diff Eq.

since for a damped particle:

$\frac{dx}{dt} = v \tag{1}$ $m \frac{dv}{dt} = -m\gamma v - \underbrace{V'(x)}_{\text{force}} + \sqrt{2\gamma k_B T m} \cdot \Gamma(t) \tag{2}$

In the limit of large damping $\gamma \gg 0$ one can neglect $\frac{dv}{dt} \cdot m \approx 0$ i.e.

$$0 = -\gamma \cdot v m + V'(x) + \sqrt{2\gamma k_B T m} \cdot \Gamma(t)$$

Thus $\frac{dx}{dt} = v = \frac{1}{\gamma m} V'(x) + \sqrt{\frac{2 k_B T}{\gamma \cdot m}} \cdot \Gamma(t)$

For $V' = mg$ $dx = dt \cdot (-g) + \frac{\Gamma(t) \cdot db \cdot D^{1/2}}{dW(t)}$

corresponding Fokker Plauk equation:

$$\partial_t P(x) = \frac{\partial}{\partial x} \cdot (g \cdot P(x)) + \frac{1}{2} D \frac{\partial^2}{\partial x^2} P_S(x) \quad \text{SMOLUCHOWSKI Equation}$$

$\Rightarrow P_S(x) = N_0 \exp(-2gx/D)$ N_0 : Normalization

Solution is normalizable only if a is finite (lower boundary)
(but b may be infinite).

Makes sense: particles keep falling down as $a \rightarrow -\infty$ but not for $b \rightarrow \infty$.

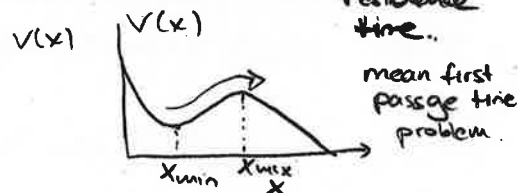
LECTURE

Non Eq. Stat Mech

Reading Material

Ch. 5.5 "Stoch. Methods" Gardiner
Ch. 4 Martin Polycopy

Fokker-Planck Equation: First passage times (and problems) / particle residence time.
eg Kramer's escape rate



Alternative treatment: see Chapter 5 Gardiner

How long does a particle stay (on average) in a certain interval $x \in [a, b]$?

The probability to find particle in $x \in [a, b]$ is:

$$\int_a^b P(x', t | x, 0) dx' \equiv G(x, t)$$

initial condition.

Let T be time where the particle escapes. Hence

Probability that $t > T$ is given also by $G(x, t)$: $\text{Prob}(T \geq t) = \int_a^b P(x', t | x, 0)$

$$\left[G(x, t): \text{probability that particle is still in interval } (a, b) \text{ at time } t \right] \iff \text{Equivalent} \left[G(x, t): \text{Probability that particle will leave } [a, b] \text{ at time } T \geq t \right]$$

Now is $G(x, t)$ related to the MEAN FIRST PASSAGE TIME (MFPT)?

$P_T(\gamma) \equiv P_T(\gamma)$

Consider: $P_T(\gamma)$ is the probability density function to have a mean passage time $\gamma \dots \gamma + d\gamma$.

Hence:

$$\left[G(x, t) = \int_t^\infty d\gamma P_T(\gamma) \right] (x) \quad \text{and} \quad \left[\langle T(x) \rangle = \int_0^\infty d\gamma \gamma P_T(\gamma) \right]$$

mean FPT.

→ Expression gives probability to exit for at $T > t$

Using (*) we can determine $P_T(\gamma)$: (reverse integration and differentiate)

$$P_T(\gamma) = - \frac{\partial G(x, t)}{\partial t}$$

And the MFPT:

$$\langle T(x) \rangle = - \int_0^\infty t \frac{\partial G(x, t)}{\partial t} dt = \int_0^\infty dt t G(x, t)$$

LECTURE

Non. Eq. Stat. Mech

Derive next ordinary differential equation for $T(x)$:

Problem: $\partial_t G(x,t)$ satisfies FORWARD F.P. equation BUT not $G(x,t) = \int_{x_1}^{x_2} p(x_1, t) dx_1$

First: $G(x,t)$ satisfies the backward Fokker Planck eqs: since: $p(x', t | x, 0) = p(x', 0 | x, -t)$

$$\partial_t p(x', t | x, 0) = A(x) \partial_x p(x', t | x, 0) + \frac{1}{2} B(x) \partial_x^2 p(x', t | x, 0)$$

and thus using $G(x,t) = \int_a^b p(x', t | x, 0) dx'$ yields:

(i.e. satisfies backward F.P.)
 $\partial_{t_0} p(x', t_0 | x, t_0) = -\partial_t p(x', t | x, t_0)$

$$\partial_t G(x,t) = A(x) \partial_x G(x,t) + \frac{1}{2} B(x) \partial_x^2 G(x,t)$$

thus for $\langle T(x) \rangle$ we obtain:

$$T(x) = \int_0^\infty G(x,t) dt$$

$$A(x) \partial_x T(x) + \frac{1}{2} B(x) \partial_x^2 T(x) = \int_0^\infty \partial_t G(x,t) dt = G(x, \infty) - G(x, 0) = -1$$

Since $B(x, \infty) = 0$: probability that particle escapes later than $T \rightarrow \infty$
 $G(x, 0) = 1$: prob. that particle escapes for T larger than 0

Thus: $A(x) \partial_x T(x) + \frac{1}{2} B(x) \partial_x^2 T(x) = -1$ Differential Eqs for TMP.

with Boundary conditions $T(a) = T(b) = 0$ since

$G(a, t) = G(b, t) = 0$ since if $x=a$ or $x=b$ particle exits immediately.
 Prob($T > t$) = 0

Thus: $T(a) = \int_0^\infty G(a, t) dt = 0 = T(b)$

Solution (no proof) for $x=a$ absorbing b. condition $\Rightarrow \partial_x G(a, t) = 0$
 $x=b$ reflecting boundary $\Rightarrow G(b, t) = 0$

$$T(x) = \int_a^b \frac{dy}{x} \frac{\psi(y)}{\psi(x)} \int_a^y \frac{\psi(z)}{B(z)} dz \quad \text{and} \quad \psi(x) = \exp \left\{ \int_a^x \frac{A(x')}{B(x')} dx' \right\}$$

$$T(a \rightarrow x) = \frac{1}{D} \int_a^x dy e^{+U(y)/D} \int_{-\infty}^y e^{-U(z)/D} dz$$

as before!

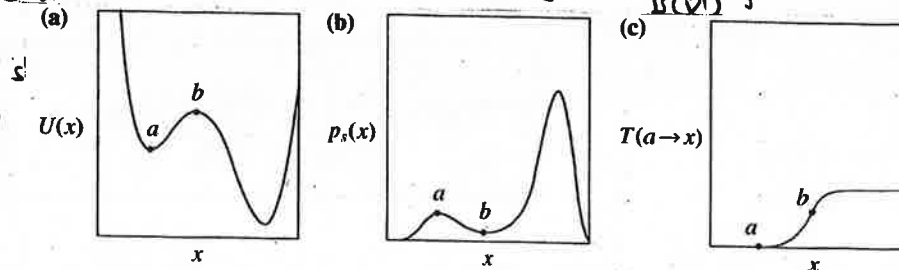


Fig. 5.3. (a) Double well potential $U(x)$; (b) Stationary distribution $p_s(x)$; (c) Mean first passage time from a to x : $T(a \rightarrow x)$